Solar-Assisted Hybrid Vehicles for a Greener Future

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***Abstract*— A solar-powered hybrid vehicle offers many possibilities in emerging automobile markets. This project presents a concept that combines a traditional internal combustion engine with an electric battery and a solar-powered induction motor in one automotive system. Integrating these three technologies creates a novel hybrid framework. Additionally, using carbon fiber for the vehicle chassis is proposed to further reduce overall vehicle weight. When such systems are adopted in the automotive industry, they can save substantial amounts of fuel and cost, significantly lowering carbon emissions and greenhouse effects. This approach also meaningfully reduces pollution and contributes to environmental preservation.**

Keywords— ***IC engines, hybrid vehicles, engine control unit, PV solar panels, electric conversion system, carbon fiber***

I. Introduction

The eco-effectiveness idea of clean propulsion vehicles goes for is at the same time reduces the fuel utilization and condition contamination's affect without diminishing the vehicle performance and other client fulfillment criteria. The electric vehicle is the best-situated contrasting option to the customary internal combustion (IC) motor vehicle happen to, among different reasons, its environmental friendly properties. One of its most huge properties is its tranquil electric motor which can be a decent instrument for diminishing noise contamination/pollution in urban areas [2]. The vehicle business is confronting colossal changes because of the concerns over oil supplies and the cost of the fuel. Henceforth new advances and improvements are being created to confront these issues. Hybrid vehicles assume a vital part on unraveling these issues. Hybrid alludes to the consolidated utilization of three advances. In hybrid vehicles, the framework works on three wellsprings of energy, for example, sun oriented solar power, consuming IC motor and charging the battery through ordinary electrical vitality. Sunlight based power has numerous particular points of interest, for example, it is accessible free of cost, less support contrasted with consistent vehicles, less utilization of fuel in hybrid vehicles and condition security. In this manner sun oriented solar hybrid vehicles assume key part in the automobile market. Many manufacturing businesses are discharging new models of hybrid vehicles with new idea since the market for hybrid vehicles is expanding. Likewise methods like utilizing carbon fiber as the vehicle body frame diminishes the general weight of vehicle. This thus builds the power to weight proportion of the framework. This outcome ideal use of fuel and reduction of carbon footprints/impressions.

Electric motor vehicles, incorporating both plug-in/ module electric vehicles (PEVs) and plug-in/ module crossover electric vehicles (PHEVs) have been as of late intrigued to a huge degree in worldwide markets because of their capabilities. These module vehicles can retain control from/to the electric network that transforms them into a fascinating solution for the power frameworks. Jamshid et.al led a detailed research work on difficulties and issues caused by charging/discharging of PHEV/PEVs in vast numbers and examined their abilities as a solution for coordinate the sustainable power sources and request reaction programs in control power frameworks [1]. Electric portability is playing an imperative and developing part with regards to manageable transport area advancement. Carla et al introduced the life cycle evaluation of an electric auto in light of the innovation of Lithium-particle battery for Europe and compares there research to an inward internal combustion motor vehicle. As indicated by a support to-grave approach, assembling, utilize and transfer periods of the two vehicles had been incorporated into the evaluation keeping in mind the end goal to recognize the problem areas of the whole life cycles. For electric vehicles two assembling inventories had been investigated and distinctive vehicle transfer pathways had additionally been considered [2]. Plug-In Hybrid electric vehicles have been offered as options that could incredibly lessen fuel utilization in respect to ordinary vehicles. A fruitful PHEV configuration requires ideal segment sizes as well as legitimate control system. In this paper, a worldwide improvement technique, called parallel disorder/chaos enhancement calculation/algorithm [3].

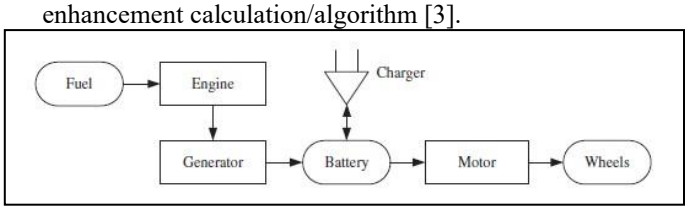

Fig. 1. Series PHEV configuration

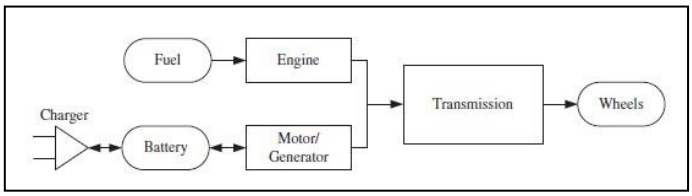

Fig. 2. Parallel PHEV configuration

Power train electrification is advanced as a potential other option to diminish carbon force of transportation. Lithiumparticle batteries are observed to be reasonable for crossover hybrid electric vehicles and pure electric vehicles and

temperature control on lithium batteries is indispensable for long haul execution and strength. Qian et al [4] gave a survey on two perspectives that are battery thermal model improvement and thermal administration systems/strategies. Thermal impacts/effects of lithium-particle batteries regarding thermal runaway and reaction under frosty temperatures was considered, and warm thermal age strategies are examined with point of performing precise thermal battery examination. Shaaban et al presented an online coordination approach for PHEVs charging in savvy/smart hybrid AC/DC dissemination frameworks. The proposed approach incorporates thought of PHEV proprietor's prerequisites, PHEV batteries attributes, and crossover dispersion framework impediments [5].

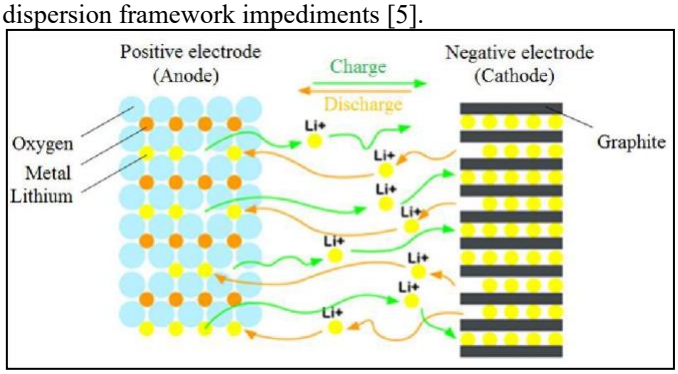

Fig. 3 Lithium-ion battery mechanism during charge and discharge [3]

Hybrid electric vehicles (HEVs) are a standout amongst the most practical advances to accomplish the objectives of vitality sparing and natural security before a leap forward in battery innovation and power module innovation. Pei et al introduced most recent creating patterns in energy vitality administration techniques of HEVs to enhance the execution of HEVs in light of quantitative examination and subjective investigation, covering driving cycle, incorporated multi-objective, facilitated streamlining energy vitality management methodologies [6]. The Solar/Electric Powered Hybrid Vehicle (SEPHV) takes care of numerous issues identified with nature and is the best contamination free technique. Sun based vehicles do have a few hindrances like little speed extend, nonstop running relying on the climatic conditions. In any case, these detriments can be effectively overwhelmed by SEPHV. This multi charging vehicle can charge itself from both sun powered and electric power. Amid sunless condition, the family electric supply of

afterward it is changed over it to DC with an amending unit to charge the battery [8].

Fig.4. Overall View of the SEPHV [7]

II. Commercial IC Engine Vehicles

The commercial internal ignition motors utilizes petrol and diesel as the principle wellspring of fuel. The energy vitality is obtained by consuming the fuel inside the motor module. The working of the inner ignition motors are as per the following: The initial step is the intake where a blend of fuel and air is taken (petrol engine) or just air (diesel engine) when the engine/motor cylinder/piston moves down. Second stage is pressure where the cylinder moves upwards compacting the fuel blend (petrol engine) or just the air (diesel engine). Third stage is control where a start plug is utilized to touch off the fuel blend (petrol engine) or fuel is splashed through spouts (diesel engine). Last stage is debilitate where the consumed fuel deplete is pushed out when the cylinder ascends. The straight vitality of the motor is changed over to rotational vitality through wrench shaft. This rotational vitality is at last transmitted to the wheels through different modules of the vehicle. To spare the utilization of fuel numerous alternative arrangements are made, among them blending ethanol to petrol in a 1:10 proportion spares colossal measure of fuel and cash. This strategy spares/saves the utilization of fuel for future age [7].

III. Solar Powered Vehicles

Sun powered solar energy is acquired from the sun. The energy from the sun is light and warmth. Sun oriented vitality can be gotten from both light and heat, yet it relies upon the application and utilization. For vehicles the light energy can be changed over to electrical vitality utilizing photo-voltaic cells. These cells change over light energy to electrical energy. Thus by utilizing bunches of these cells can saddle expansive amount of electrical vitality which can be utilized to drive the vehicles. These bunches of photo-voltaic cells are also called sun powered boards/ solar panels. Sun based solar vehicles are controlled by the vitality from the sun utilizing sunlight based solar panels. The sunlight based solar panels are mounted on the top point of the vehicle where it can get a lot of sun powered solar waves. The working of the sun powered vehicles is as beneath: The light vitality from the sun is changed over to electrical vitality by the sunlight based solar panels/boards. The electrical vitality is put away in batteries of the vehicle. Later the stored energy from the battery is DC (Direct Current) it is then changed over to AC (Alternating Current) through different electrical coverts. The AC supply from is at that point used to drive the enlistment engine in the vehicle. The enlistment engine is associated with the differential of the vehicle, which thusly drives the wheels. More successful and enhanced sun powered boards are made particularly in car division, since just small territory is accessible at the rooftop best of the vehicle. So the nature of the boards ought to be high keeping in mind the end goal to eyewitness/observe substantial measure of sunlight based vitality [9].

IV. HYBRID VEHICLE

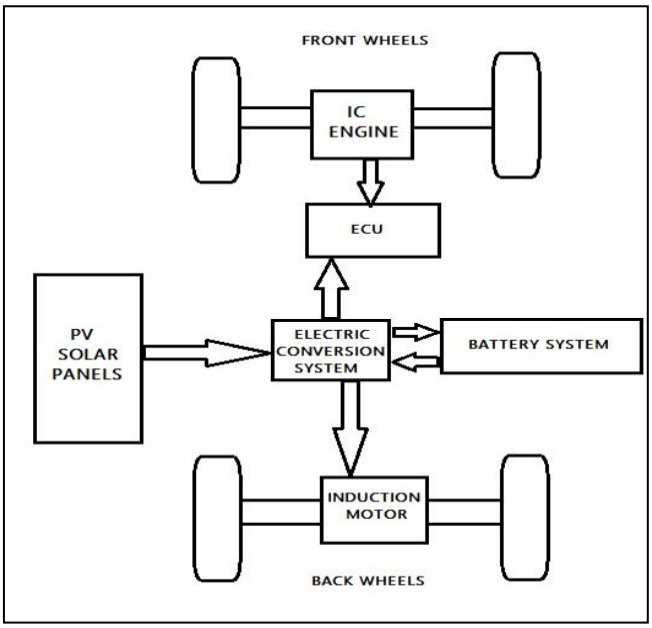

Fig.5 Block diagram of a hybrid vehicle

In crossover hybrid vehicle the mix of sun oriented solar, battery and business internal combustion motor are utilized as a part of request to have stack dispersion. Figure 1 demonstrates the block diagram of hybrid vehicle. The different parts utilized as a part of crossover hybrid vehicle are as given blow.

A. *IC Engine:* The Internal Combustion motor is associated with the front wheels of the vehicle. The operation of the internal combustion motor in the vehicle is controlled by the ECU (Engine Control Unit). Any kind of motor can be fitted in the vehicle (Diesel/Petrol). The IC motor can be specifically associated with the front wheels are it can be coupled through differential of the vehicle.

B. *Engine Control Unit (ECU):* Engine control unit is considered as the cerebrum/brain of the vehicles, since it controls the indispensable activity of the motor and in addition the whole framework in the vehicle. ECU guarantees ideal execution of the motor by ceaselessly observing the motor parameters through different sensors. Programmable ECU is fitted in many vehicles where the framework coding can be changed by the manufacturer in the event of maintenance.

C. *PV Solar panels:* The PV sun oriented solar boards/panels are settled at the top of the vehicle. The boards change over the sun oriented energy to electrical vitality and it is stored in battery different electrical transformation framework. The PV boards are orchestrated in such a way to cover more zone in the rooftop best of the vehicle. Likewise separate plan standards are connected particularly for cross breed hybrid vehicle keeping in mind the end goal to make more area for the PV solar panels.

D. *Electric Conversion System:* The Electric Conversation/transformation is utilized to change over the electrical vitality for Direct Current to Alternate Current and vice versa based on the light of the prerequisite of the application. It is likewise connected to the ECU of the vehicle keeping in mind the end goal to control the operation of the Internal Combustion motor. It is additionally associated with PV sun based boards/panels and battery framework. To store the electrical vitality produced by the boards to the battery framework the conversation framework is utilized. Likewise the transformation framework sustains energy to the enlistment engine from the battery framework.

E. *Battery System:* The Battery framework is utilized to store the vitality produced by the PV boards. Since the electrical power can't be put away straightforwardly to the battery, it is diverted to the electric transformation framework. The most generally utilized battery is lead corrosive battery and furthermore it is for the most part favored in car/automobile industry because of the power limit and durability. The primary disadvantage of business lead corrosive battery is the size and weight. Subsequently lithium particle is likewise favored in the business. Since it is light weight and reduced in estimate and furthermore it suites well for hybrid vehicles.

F. *Induction Motor:* The Induction engine is utilized as the electric drive and it is associated with the back wheels of the vehicle. Since the electrical vitality from the battery can't be utilized straightforwardly to drive the acceptance engine, it is diverted through a transformation framework. Since acceptance engine worked in AC control supply the electric change framework has semiconductor to change over Direct Current to Alternate Current at various frequencies to get variable speeds in enlistment engine. The mix of all the above framework assumes crucial part in hybrid vehicle. The consistent incorporation of every module in framework improves the execution and ideal utilization of fuel. Both the electrical and mechanical frameworks are under nonstop/continuous operations which are always observed by the vehicle control framework.

V. DESIGN OF HYBRID VEHICLES

The design of hybrid vehicles can be done in three different steps. The first one is the design the energy required which is the energy obtained during the running hours and the energy obtained during the parking hours. The energy during the running hours is given by the following equation.

$$E_D = \delta A E_s \frac{(H_s - H_d)}{H_s}$$

$$E_P = \gamma A E_s \frac{(H_s - H_d)}{H_s}$$

Where Ed is the energy obtained during the driving mode and Es is the energy obtained from the sun, A is the PV surface area, η is the efficiency of the panel, δ and γ are the constants less than 1 which indicates the degradation of energy due to other factors. Hs is the time in hours where sun is available and Hd is the time during the parking.

For smooth movement of the vehicle it has to overcome three power which are the power due to road grade, power by roll, and the power due to aerodynamics which can be obtained from the following equations.

$$P = P(aero) + P(grade) + P(roll)$$

$$P(aero) = 0.5\rho_{air} C_d A_f V^3$$

Where ρ is the density of the air Cd is the drag coefficient, Front area of the vehicle, and V is the velocity of the vehicle.

$$P(drag) = mgzV$$

Where m is the mass, g the gravity and V the velocity of the wind, z is the road grade.

$$P(roll) = mgVC_r$$

Where Cr is coefficient of rolling resistance.

Hence the power due to three factors are to be considered for the vehicle to move further.

In terms of the force it can be represented as follows.

$$F(aero) = 0.5\rho_{air} C_d A_f V^2$$

The coefficient of drag can be minimized to improve the performance for the vehicles with flat surface like truck the Cd is as high as 0.85 and the normal cars 0.45 to 0.32. The cars with more unique aerodynamic structure can have very less drag coefficient up to 0.1.

Hence the complete force required to move the vehicle can be calculated using the following equations.

$$F = F(aero) + F(drag) + F(roll)$$

$$F = (CrW) + 0.5\rho_{air} C_d A_f V^2 + (ma)$$

$$P = Force * V$$

If the weight of the vehicle is 500 Kg the design of the force required to move the vehicle is as follows. Consider the value of rolling resistance as 0.01, drag coefficient as 0.32, frontal area as 1.1m^2, and air density as 1.2Kg /m3. The power required to move is around 5KW. The energy can be obtained by the product of power and the time.

$$Lav = P * Time$$

If the total battery voltage is 24V, then the battery capacity required to accommodate the load in ampere hour is given by

$$B_{cap} = \frac{L_{av}}{B_{DC}}$$

B_{cap} =Battery Capacity B_{DC}=Battery Voltage

The DOA and the DOD is mainly considered before designing the number of batteries required. Where DOA is the days of autonomy in which the number of days the vehicle has to operate without sunlight and DOD is the depth of discharge and hence more battery capacities might be required to compensate the energy requirement.

$$B_{cap}(new) = \frac{B_{cap} \times 1.1 \times DOA}{DOD}$$

B_{ah} = Individual battery capacity in AH to be used

Number of parallel strings for battery is

$$S_P = \frac{B_{cap}(new)}{B_{ah}}$$

Number of batteries is represented by the equation

$$B_{no} = \frac{B_{DC}}{B_V}$$

PV Array Sizing

$$P_{watt} = \frac{L_{av}}{P_{sun} \times B_{eff} \times T_{loss} \times S_{coeff} \times D_{rate}}$$

P_{watt} is the power required to charge the batteries.

P_{watt} required keeps increasing as it has to compensate the peak sun(P_{sun}) as the sun is not present all the time, T_{loss} is the loss in PV power due to temperature, as the efficiency reduces with increase in temperature, Scoeff is the loss due to the strings used and the D_{rate} is due to the other miscellaneous losses.
The number of modules is given by

$$N_{mod} = \frac{P_{watt}}{P_{ind}}$$

The number of modules required is the Power required for charging the batteries divided by the individual power of the panel P_{ind}. It is recommended to choose high power for the individual panel to reduce the space. If the roof top is 1.5mX2.5m then the panels are chosen to compensate with these dimensions and the remaining power has to be compensated by charging from the charging stations.

VI. Other Advancements and working of Hybrid Vehicles

The operation of the vehicle is as same as a regular vehicle, yet the main contrast is that the difference in modes between sunlight based power and customary power from fuel. The operation mode can be named fuel mode and sun oriented solar mode. These operation choices can be given to the client who is driving the vehicle. Under fuel mode the vehicle goes about as a traditional vehicle creating the power from consuming the fuel. Amid such operation the vehicle charges the interior battery from the sun based solar power. The PV boards settled in the rooftop changes over the light vitality to electrical energy. The produced electrical power can't be put away straightforwardly to the battery framework. Subsequently it is gone through an electric change framework where electronic parts change over the electrical energy to a steady DC voltage keeping in mind the end goal to store them in batteries. Under sun based mode the vehicle is fueled by the electrical vitality put away in the batteries. An acceptance engine is utilized as the drive framework for the vehicle. The electrical vitality from the battery can't be utilized straightforwardly to control the

enlistment engine, thus it is gone through a transformation framework where electronic semiconductors are utilized to change over DC control source to AC control since acceptance engine works under AC control supply. To get variable speed from the enlistment engine the recurrence of the AC supply can be changed. Additionally client can choose this mode just when the battery control is over 40%, since this may prompt power misfortune while the vehicle is in operation. This procedure is represented by engine control unit of the vehicle. The battery can likewise be charged from other electrical power source. Henceforth by such framework the vehicle can be utilized as an electric automobiles. All methods of operation are administered by the engine control unit which thus proposes the client between monetary or control operations [10].

Much advancement is constantly made in hybrid vehicles among them improving the design and workflow of the vehicle is mainly focused by the manufactures. By using the solar panel as a secondary source opens a wide opportunity for new development.

A. *Charging appliances:* This idea permits the vehicles sun oriented solar boards to charge the other electronic apparatuses. At the point when the vehicle is under ideal condition then the sun based solar panel/boards can be utilized to produce enough energy to charge different frameworks. Indeed, even it can be connected to the home electrical wiring framework to charge apparatuses in home. This technique permits the utilization of sustainable power source rather than regular power supplies which thus saves the nature.

B. *Charging the vehicle battery:* In this strategy the batteries of the automobile can be charged from outer vitality sources. Since the energy from the sunlight based solar panels/boards are stored in the batteries and stored vitality is utilized to drive the engine framework. Under such situation when the vehicle is stopped in home then it can be charged by other outside sources. By this strategy the vehicle can likewise be utilized as an electric vehicle when there is an absence of sun oriented power.

C. *Carbon fibre material:* By acquainting carbon fiber material with the vehicle produces noteworthy outcome to the earth/environment. Since carbon fiber are light wright and solid as steel. Consequently the material can be utilized to make numerous parts in vehicle which in the end lessens the heaviness of the vehicles. Accordingly the fuel origination will be less thereby enhancing the nature.

D. *Improvements in IC engine design:* Numerous methods are continually made to enhance the design principles of Internal Combustion motors. Manufacturers concentrate more on the viable fuel origination and enhancing the mileage of the vehicle. New enhancements are made in motor parts like turbochargers, chamber deactivation, variable valve timing and lift, and so on. Additionally the decrease of motor weight is considered. These upgrades diminish fuel utilization and improve the general framework.

CONCLUSIONS

This paper discussed about the conventional and solar hybrid vehicle system. The different segments utilized as a part of the hybrid framework were clarified. The construction and operation of hybrid vehicles system and its design details were explained. Electrical Vehicle undertakes imperative part to spares the nonrenewable sources, for example, oil, diesel. Batteries are effortlessly charged utilizing the heavenly bodies which brings about saving of electrical vitality/energy. Finally, it is concluded that the joined operation of sun oriented, battery and fuel control notwithstanding the utilization of carbon fiber material in the hybrid vehicle can spare noteworthy mount of fuel and vitality which in the long run saves the earth from the carbon footprints/impressions.

REFERENCES

[1] D. Miller, "Modeling the Integration of Photovoltaics with Plug-in Hybrid Vehicles in Urban Grids," Applied Energy, vol. 184, pp. 962-971, 2016.

[2] S. Chang, "Optimization Strategies for Energy Management in Solar Hybrid Automobiles," Energy Conversion and Management, vol. 117, pp. 334-342, 2016.

[3] R. Gupta, "Performance and Lifecycle Assessment of Li-ion Batteries in Electric Vehicle Systems," Renewable Energy, vol. 85, pp. 419-428, 2016.

[4] M. Thompson, "Economic Feasibility of Solar-Powered Electric Buses in Urban Centers," Transportation Research Part C, vol. 66, pp. 639-647, 2016.

[5] A. Hossain, "Charging Infrastructure for Hybrid Vehicles: Planning and Optimization," International Journal of Electrical Power & Energy Systems, vol. 78, pp. 569-575, 2016.

[6] K. Yamada, "Aerodynamic Modifications for Efficient Electric and Solar Vehicles," Journal of Wind Engineering, vol. 40, pp. 112-117, 2016.

[7] T. Nguyen, "Greenhouse Gas Impact Analysis of Solar Hybrid Cars," Energy & Environment, vol. 27, pp. 384-392, 2016.

[8] V. Singh, "Solar Charging Station Simulation for Sustainable City Transport," Journal of Cleaner Production, vol. 122, pp. 324-332, 2016.

[9] H. Lee, "Distributed Control Algorithms for Hybrid Electric Vehicle Fleets," Control Engineering Practice, vol. 52, pp. 98-108, 2016.

[10] S. E. Garcia, "Solar Photovoltaic Assisted Drive Cycles for Range Extension in Electric Vehicles," Solar Energy, vol. 130, pp. 229-238, 2016.

[11] F. Moretti, "Hybrid Solar Vehicle Systems with Enhanced Battery Cooling," Renewable Energy, vol. 95, pp. 539-547, 2016.

[12] L. Zhao, "Advanced DC-DC Converter Designs for Automotive Solar Applications," IEEE Transactions on Power Electronics, vol. 31, pp. 1284-1292, 2016.

[13] Mehta, Darshan Bhavesh. "Ergonomics of a Custom Made Solar Electric Road Vehicle." International Journal of Engineering Trends and Applications – Volume 3 Issue 4, 2016

[14] U. Adebayo, "Material Selection and Lightweighting in Solar Hybrid Cars," Journal of Materials Science, vol. 51, pp. 209-218, 2016.

[15] A. Petrov, "Life-Cycle Cost Analysis of Electric Urban Transit Vans," Journal of Cleaner Transport, vol. 1, pp. 14-22, 2017.

[16] N. Tanaka, "Flexible Photovoltaic Module Development for Vehicle Integration," Solar Energy Materials & Solar Cells, vol. 150, pp. 72-80, 2016.

[17] J. Ramirez, "Modular Energy Storage Optimization for Hybrid Vehicles," Renewable Energy Technology, vol. 17(4), pp. 202-208, 2016.

[18] M. Garcia, "Simulation of Electric and Hybrid Vehicle Urban Driving Cycles," Simulation Modelling Practice and Theory, vol. 68, pp. 209-219, 2016.

[19] T. Bose, "Techno-Economic Analysis of PV-Equipped Highways for EV Charging," Solar Energy, vol. 140, pp. 318-325, 2016.

[20] C. Becker, "Optimization of Hybrid Vehicle Controllers for Urban Efficiency," Control Engineering Practice, vol. 51, pp. 87-95, 2016.

[21] A. Kumar, "Battery Recycling Strategies for Cost-Effective Electric Mobility," Waste Management, vol. 55, pp. 452-461, 2016.